Patient-Independent Multi-Horizon EEG Seizure Prediction: An Event-Level Evaluation of Generalization and Alarm Reliability

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Abstract—Reliable seizure forecasting requires evaluation on patients excluded from model development and alarm thresholds that remain useful at low false-alarm rates. This project develops a shared multi-horizon model for scalp EEG that estimates preictal risk at 5-, 10-, 15- and 30-minute seizure-onset horizons with a fixed 5-minute seizure-prediction horizon. The pipeline filters and resamples EEG, maps recordings to a canonical bipolar montage, and forms overlapping windows. Temporal CNN, spectral band-power and time-frequency STFT branches are fused with attention; an adversarial patient-identity branch is used to encourage patient-independent representations. Experiments use a locally available CHB-MIT subset of 16 case folders representing 15 people, with patient-wise LOSO splits, validation-only threshold selection, and matched CNN, LSTM and CNN-LSTM baselines plus staged ablations. Performance is reported using event sensitivity, false alarms per hour and warning lead time, with person-level uncertainty. At the 30-minute horizon and 0.5 false-alarm/hour validation target, the full model achieved 78.3% ± 35.2% mean event sensitivity and 0.81 ± 1.44 observed false alarms/hour across 15 people; 10 of 15 met the target. On Siena, a single frozen CHB-MIT fold detected 45 of 47 events at this operating point but produced 9.13 false alarms/hour, showing poor threshold transfer. The work provides a controlled event-level comparison and an external transfer assessment; it is retrospective and is not clinically validated.

Keywords—EEG, seizure prediction, multi-horizon learning, leave-one-subject-out evaluation, event sensitivity, false alarms per hour, explainable AI.